

Hogfish

Species: *Lachnolaimus maximus*, Stock code: Hogfish - Florida Keys - East Florida

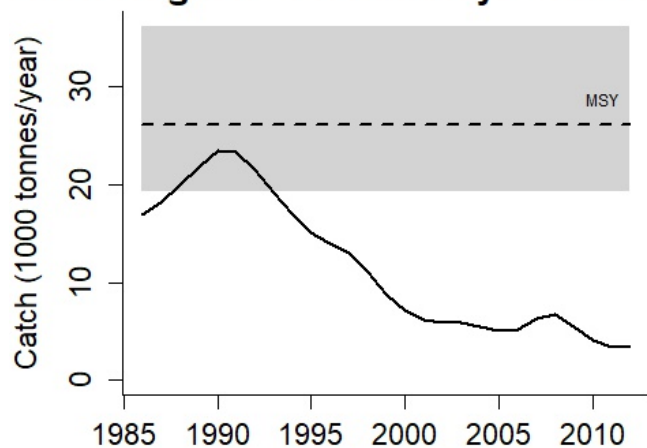
Region: Southeast Shelf

Marine Ecoregion: Florida Keys - East Florida

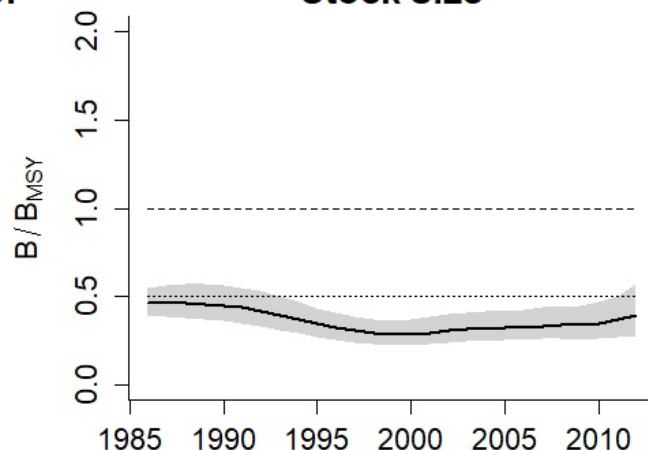
Reconstructed catch data used from years 1986 - 2012

For figure captions and method see <http://www.seaaroundus.org/cmsy-method>

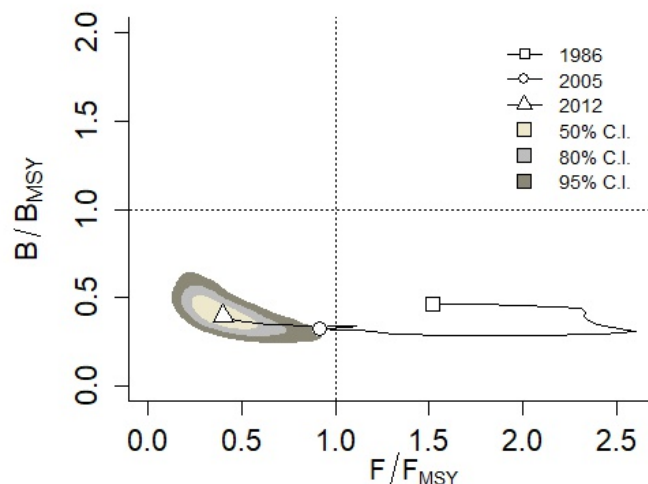
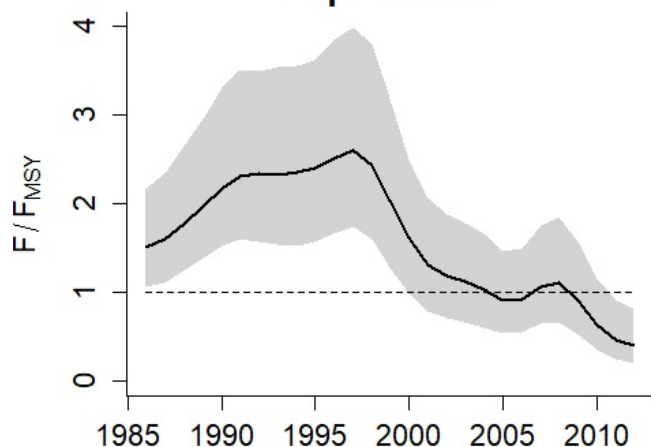
Catch Hogfish - Florida Keys - East Floi



Stock size



Exploitation



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Results for management (based on BSM analysis)

$F_{msy} = 0.0927$, 95% CL = 0.0522 - 0.139 (if $B > 1/2 B_{msy}$ then $F_{msy} = 0.5 r$)

$F_{msy} = 0.0739$, 95% CL = 0.0416 - 0.111 (r and F_{msy} are linearly reduced if $B < 1/2 B_{msy}$)

$MSY = 26.1$, 95% CL = 19.3 - 36.2; $B_{msy} = 284$, 95% CL = 180 - 498 (1000 tonnes)

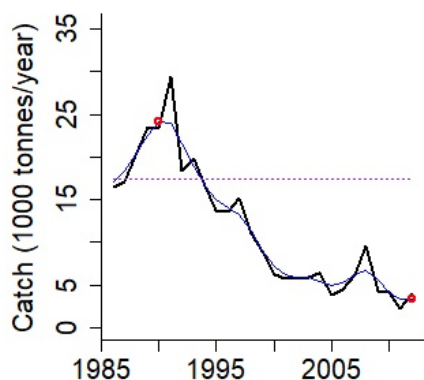
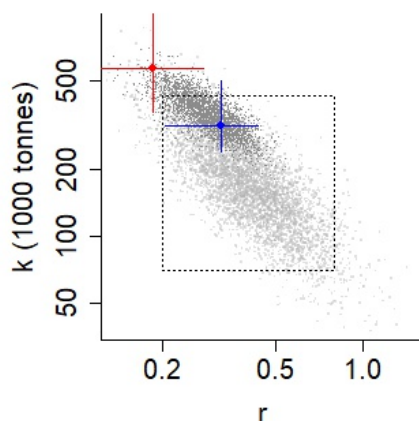
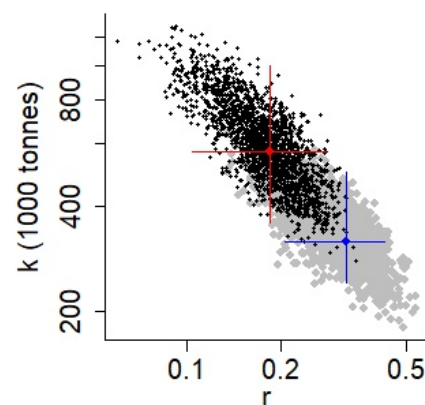
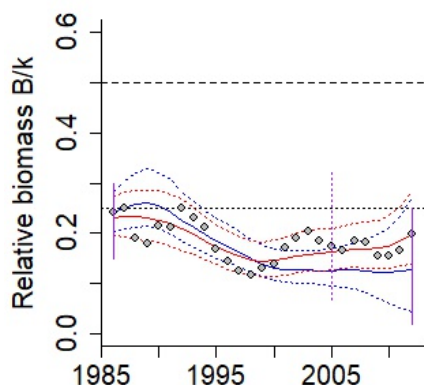
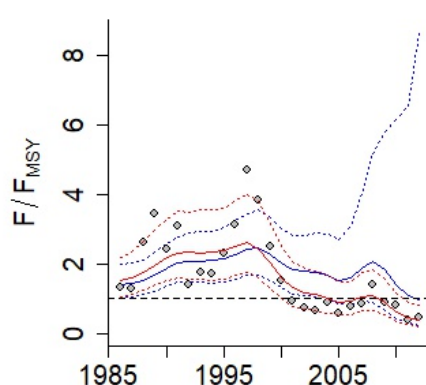
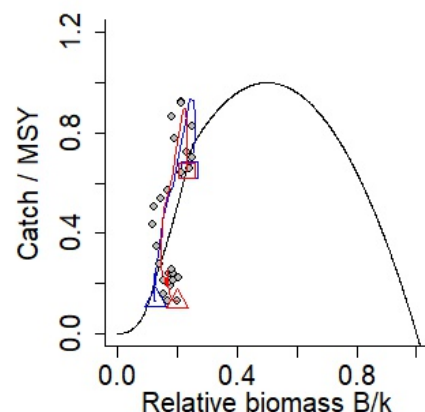
Biomass in last year = 114, 95% CL = 71.4 - 194 (1000 tonnes)

B/B_{msy} in last year = 0.398, 95% CL = 0.279 - 0.571

Fishing mortality in last year = 0.0291, 95% CL = 0.0157 - 0.051

$F/F_{msy} = 0.4$, 95% CL = 0.208 - 0.828

Comment: NOAA Assessment model SS3. RF OK

A: Hogfish - Florida Keys - East Flor**B: Finding viable r-k****C: Analysis of viable r-k****D: Stock size****E: Exploitation rate****F: Equilibrium curve**

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Results of CMSY analysis conducted in JAGS

$r = 0.322$, 95% CL = 0.206 - 0.429; $k = 315$, 95% CL = 241 - 496 (1000 tonnes)

MSY = 25.4, 95% CL = 19.7 - 33.4 (1000 tonnes/year)

Relative biomass last year = 0.129 k , 95% CL = 0.0443 - 0.271

Exploitation $F/(r/2)$ in last year = 1.11

Results from Bayesian Schaefer model using catch and CPUE

$r = 0.185$, 95% CL = 0.104 - 0.279; $k = 569$, 95% CL = 359 - 996

r - k log correlation = -0.813

MSY = 26.1, 95% CL = 19.3 - 36.2 (1000 tonnes/year)

Relative biomass in last year = 0.129 k , 95% CL = 0.0443 - 0.271

Exploitation $F/(r/2)$ in last year = 0.582

$q = 3.53$, 95% CL = 2.07 - 5.27

Prior range of q = 1.16 - 28.4

Relative abundance data type = CPUE

Prior initial relative biomass = 0.15 - 0.3 expert

Prior intermediate relative biomass = 0.0678 - 0.323 in year 2005 default

Prior final relative biomass = 0.02 - 0.25 expert

Prior range for r = 0.2 - 0.8 default, prior range for k = 69.7 - 427 (1000 tonnes) default

Source for relative biomass:

NOAA Fisheries. (2020). Assessment Summary Data. Retrieved from www.st.nmfs.noaa.gov/stocksmart. 12/13/2020